Hardy-Weinberg + Random Fitness: Allele Frequency Dynamics

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2026-05-29

Contents

Overview

This project simulates **allele frequency dynamics** under the Hardy-Weinberg equilibrium framework, incorporating:

- **Viability selection** with randomly assigned genotype fitnesses
- **Genetic drift** via the Wright-Fisher binomial sampling model
- **Scaling analysis** across population sizes from N = 500 to N = 100,000

The simulation replicates and extends the results of:

> Lewontin, R.C., Ginzburg, L.R., & Tuljapurkar, S.D. (1978). Heterosis as an explanation for large amounts of genic polymorphism. *Genetics*, 88(1), 149–169.

The core biological question: **when does nonadaptive selection maintain heterosis (heterozygote advantage), and how does population size determine whether a stable polymorphic equilibrium can be reached?**

Part 1: Lewontin (1978) — Multi-Allele Simulation

Background

With two alleles (AA, AB, BB), a stable polymorphic equilibrium exists when the heterozygote AB has higher fitness than both homozygotes. By simple probability, this occurs in **1/3 of random fitness matrices**. Lewontin et al. showed this proportion collapses toward zero as allele number increases.

Functions

```
library(ggplot2)
library(dplyr)
library(scales)

set.seed(1978) # Lewontin et al. year

# Check for stable interior equilibrium in an n-allele fitness matrix
check_stable_equilibrium_multiallele <- function(W) {
  n <- nrow(W)
  A2 <- W
  for (i in 2:n) {
    A2[i, ] <- W[i, ] - W[1, ]
  }
  A2[1, ] <- 1
  b2 <- c(1, rep(0, n - 1))

  tryCatch({
    p_star   <- solve(A2, b2)
    feasible <- all(p_star > 0)
    if (!feasible) return(FALSE)
```

```r
    marginal     <- W %*% p_star
    equal_fitness <- (max(marginal) - min(marginal)) < 1e-8
    return(feasible && equal_fitness)
  }, error = function(e) FALSE)
}

# Simulate proportion of stable equilibria for n alleles
simulate_multiallele <- function(n_alleles_vec = 2:6,
                                 n_matrices = 50000) {
  results_df <- data.frame()
  for (n in n_alleles_vec) {
    stable_count <- 0
    for (i in seq_len(n_matrices)) {
      W <- matrix(0, n, n)
      for (row in 1:n) {
        for (col in row:n) {
          val <- runif(1)
          W[row, col] <- val
          W[col, row] <- val
        }
      }
      if (check_stable_equilibrium_multiallele(W)) {
        stable_count <- stable_count + 1
      }
    }
    prop <- stable_count / n_matrices
    results_df <- rbind(results_df,
                        data.frame(n_alleles  = n,
                                   proportion = prop,
                                   n_matrices = n_matrices))
  }
  return(results_df)
}
```

Run Simulation

```r
# 2-allele case: verify ~1/3 theoretical result
two_allele_results <- replicate(50000, {
  f <- runif(3)        # AA, AB, BB
  (f[2] > f[1]) & (f[2] > f[3])
})
p2 <- mean(two_allele_results)
cat(sprintf("2-allele stable proportion: %.4f (theoretical: 0.3333)\n", p2))
```

```
## 2-allele stable proportion: 0.3313 (theoretical: 0.3333)
```

```r
# Multi-allele sweep
lewontin_results <- simulate_multiallele(
  n_alleles_vec = 2:6,
  n_matrices    = 50000
```

```
)
print(lewontin_results)
```

```
##   n_alleles proportion n_matrices
## 1         2    0.66674      50000
## 2         3    0.41404      50000
## 3         4    0.22534      50000
## 4         5    0.12602      50000
## 5         6    0.06388      50000
```

Plot: Replicating Figure 4.1

```
ggplot(lewontin_results, aes(x = n_alleles, y = proportion)) +
  geom_line(color = "#2c7bb6", linewidth = 1.2) +
  geom_point(color = "#2c7bb6", size = 3.5) +
  geom_hline(yintercept = 1/3, linetype = "dashed",
             color = "gray50", linewidth = 0.8) +
  annotate("text", x = 2.4, y = 0.355,
           label = "Theoretical 1/3 (2 alleles)",
           color = "gray40", size = 3.5) +
  scale_x_continuous(breaks = 2:6) +
  scale_y_continuous(labels = percent_format(),
                     limits = c(0, 0.4)) +
  labs(
    title    = "Nonadaptive Selection & Heterosis",
    subtitle = "Proportion of random fitness matrices with stable polymorphic equilibrium\n(Lewontin, Gi
    x        = "Number of Alleles",
    y        = "Proportion of Stable Equilibria",
    caption  = sprintf("Each point: %s random symmetric fitness matrices",
                       comma(50000))
  ) +
  theme_minimal(base_size = 13) +
  theme(plot.title       = element_text(face = "bold"),
        panel.grid.minor = element_blank())
```

Part 2: Wright-Fisher Simulation — N = 1,000

Background

We simulate allele frequency trajectories using the **Wright-Fisher model** with viability selection. At each generation:

1. HWE genotype frequencies: $f_{AA} = p^2$, $f_{Aa} = 2pq$, $f_{aa} = q^2$
2. Selection shifts allele frequency:
$$p' = \frac{p^2 w_{AA} + pq \cdot w_{Aa}}{\bar{w}}$$
3. Genetic drift: $p_{t+1} \sim \text{Binomial}(2N,\ p')/2N$

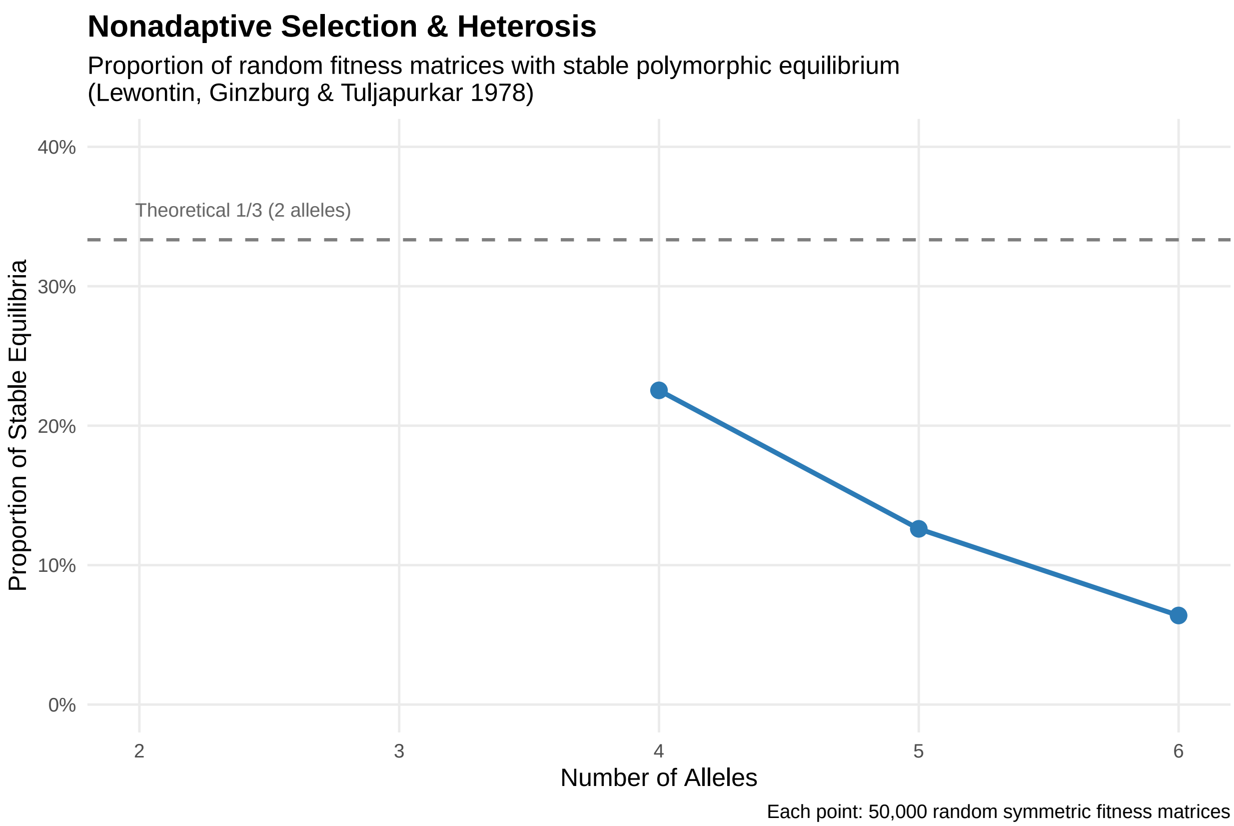

Figure 1: Proportion of random fitness matrices with stable polymorphic equilibrium, by number of alleles. Replicates Lewontin et al. (1978).

Simulation Function

```
simulate_allele_freq <- function(N, n_gen, p0, rep_id,
                                 stable_window    = 20,
                                 stable_threshold = 0.01) {
  w_AA    <- runif(1)
  w_Aa    <- runif(1)
  w_aa    <- runif(1)
  het_adv <- (w_Aa > w_AA) & (w_Aa > w_aa)

  freq    <- numeric(n_gen)
  freq[1] <- p0
  p       <- p0

  for (g in 2:n_gen) {
    q       <- 1 - p
    w_bar   <- p^2 * w_AA + 2*p*q * w_Aa + q^2 * w_aa
    p_prime <- (p^2 * w_AA + p*q * w_Aa) / w_bar
    p       <- rbinom(1, 2 * N, p_prime) / (2 * N)
    freq[g] <- p
    if (p == 0 | p == 1) { freq[g:n_gen] <- p; break }
  }

  final_freqs <- freq[(n_gen - stable_window + 1):n_gen]
  final_p     <- freq[n_gen]
  range_final <- max(final_freqs) - min(final_freqs)

  if      (final_p >= 0.99)                  outcome <- "Fixed (A)"
  else if (final_p <= 0.01)                  outcome <- "Lost (A)"
  else if (range_final < stable_threshold)   outcome <- "Stable Polymorphism"
  else                                       outcome <- "Unstable / Drifting"

  data.frame(
    generation = 1:n_gen,
    frequency  = freq,
    replicate  = rep_id,
    w_AA       = round(w_AA, 3),
    w_Aa       = round(w_Aa, 3),
    w_aa       = round(w_aa, 3),
    het_adv    = het_adv,
    outcome    = outcome,
    stringsAsFactors = FALSE
  )
}

outcome_colors <- c(
  "Fixed (A)"           = "#e41a1c",
  "Lost (A)"            = "#377eb8",
  "Stable Polymorphism" = "#4daf4a",
  "Unstable / Drifting" = "#ff7f00"
)
```

Run: N = 1,000 | 150 Generations

```r
set.seed(42)
N <- 1000; n_gen <- 150; n_reps <- 50; p0 <- 0.5

data_1000 <- data.frame()
for (rep_id in 1:n_reps) {
  data_1000 <- rbind(data_1000,
                     simulate_allele_freq(N, n_gen, p0, rep_id))
}

summary_1000 <- data_1000 %>%
  group_by(replicate, outcome, w_AA, w_Aa, w_aa, het_adv) %>%
  summarise(final_freq = last(frequency), .groups = "drop")

cat("=== N = 1,000 Outcomes ===\n")
```

```
## === N = 1,000 Outcomes ===
```

```r
print(table(summary_1000$outcome))
```

```
##
##          Fixed (A)           Lost (A) Unstable / Drifting
##                 13                 21                  16
```

Plot: Trajectories — N = 1,000

```r
ggplot(data_1000,
       aes(x = generation, y = frequency,
           group = replicate, color = outcome)) +
  geom_line(alpha = 0.55, linewidth = 0.5) +
  geom_hline(yintercept = 0.5, linetype = "dashed",
             color = "grey50", linewidth = 0.5) +
  scale_color_manual(values = outcome_colors) +
  scale_y_continuous(limits = c(0, 1),
                     labels = percent_format(accuracy = 1)) +
  labs(
    title    = "Allele A Frequency Over Generations (N = 1,000)",
    subtitle = paste0(n_reps, " replicates  |  ", n_gen,
                      " generations  |  Random fitness each run"),
    x = "Generation", y = "Frequency of Allele A", color = "Outcome",
    caption = "Wright-Fisher model | HWE + viability selection + genetic drift"
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position  = "bottom",
        plot.title       = element_text(face = "bold"),
        panel.grid.minor = element_blank())
```

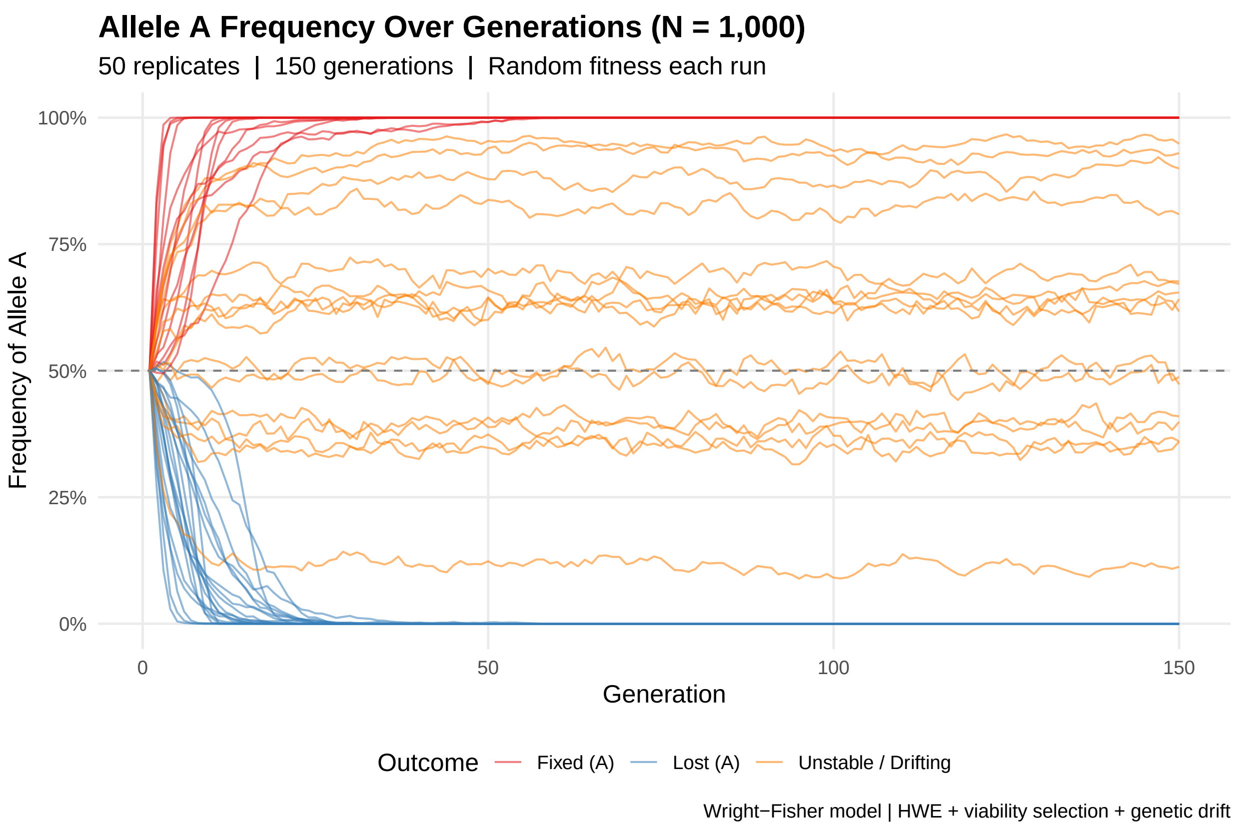

Figure 2: Allele A frequency trajectories for 50 replicates with N=1,000. Rapid fixation or loss dominates due to strong genetic drift.

Part 3: Extended Simulation — N = 10,000 | 500 Generations

Run

```r
set.seed(42)
N <- 10000; n_gen <- 500

data_10000 <- data.frame()
for (rep_id in 1:n_reps) {
  data_10000 <- rbind(data_10000,
    simulate_allele_freq(N, n_gen, p0, rep_id,
                         stable_window = 50))
}

summary_10000 <- data_10000 %>%
  group_by(replicate, outcome, w_AA, w_Aa, w_aa, het_adv) %>%
  summarise(final_freq = last(frequency), .groups = "drop")

cat("=== N = 10,000 Outcomes ===\n")
```

```
## === N = 10,000 Outcomes ===
```

```r
print(table(summary_10000$outcome))
```

```
##
##           Fixed (A)            Lost (A) Stable Polymorphism Unstable / Drifting
##                  18                  18                   1                  13
```

Plot: Trajectories — N = 10,000

```r
ggplot(data_10000,
       aes(x = generation, y = frequency,
           group = replicate, color = outcome)) +
  geom_line(alpha = 0.55, linewidth = 0.5) +
  geom_hline(yintercept = 0.5, linetype = "dashed",
             color = "grey50", linewidth = 0.5) +
  scale_color_manual(values = outcome_colors) +
  scale_y_continuous(limits = c(0, 1),
                     labels = percent_format(accuracy = 1)) +
  labs(
    title    = "Allele A Frequency Over Generations (N = 10,000)",
    subtitle = paste0(n_reps, " replicates  |  ", n_gen,
                      " generations  |  Random fitness each run"),
    x = "Generation", y = "Frequency of Allele A", color = "Outcome",
    caption = "Wright-Fisher model | HWE + viability selection + genetic drift"
  ) +
  theme_minimal(base_size = 13) +
  theme(legend.position  = "bottom",
        plot.title       = element_text(face = "bold"),
        panel.grid.minor = element_blank())
```

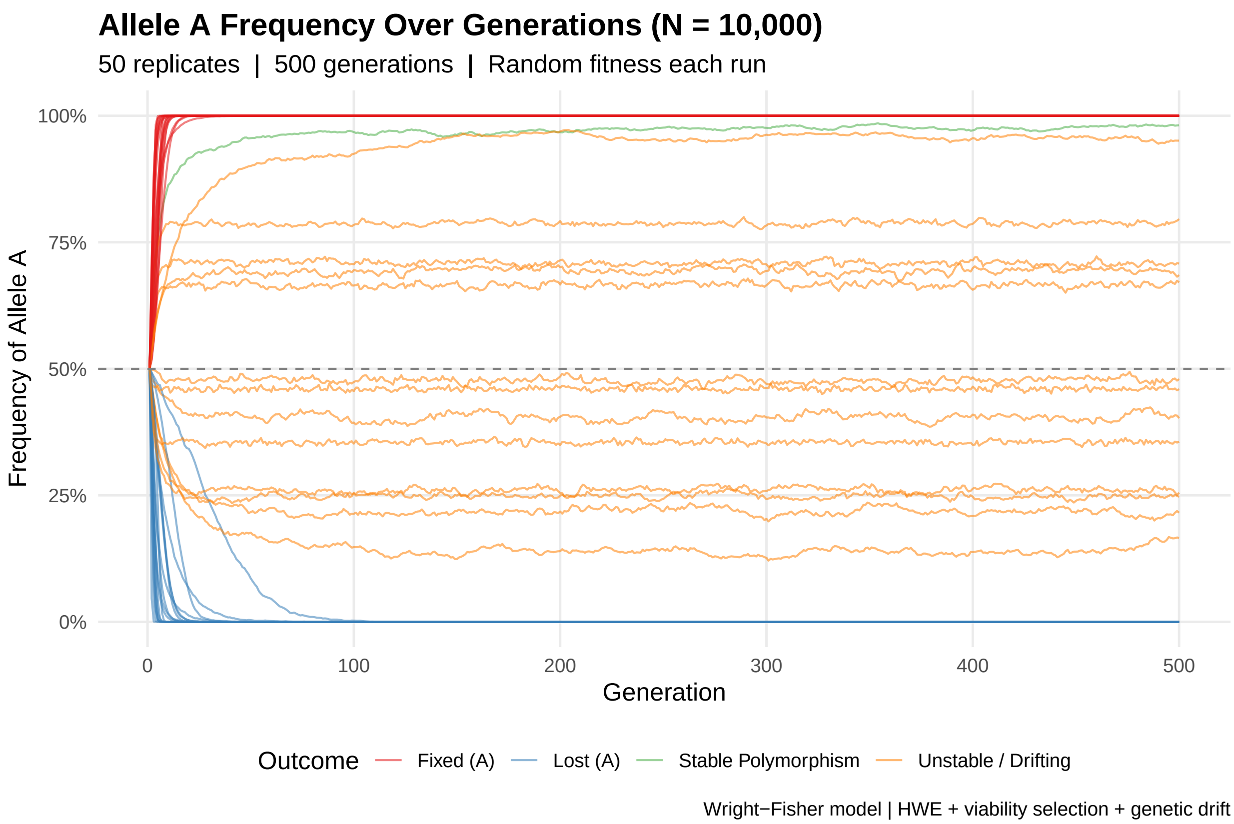

Figure 3: Allele A frequency trajectories for N=10,000. Stable polymorphism (green) now emerges as drift is reduced.

Plot: Stable Polymorphism Cases with Theoretical p*

```
stable_ids  <- summary_10000$replicate[
  summary_10000$outcome == "Stable Polymorphism"]

if (length(stable_ids) > 0) {
  stable_data <- data_10000[data_10000$replicate %in% stable_ids, ]
  stable_data$label  <- paste0("Rep", stable_data$replicate,
                               " | AA=", stable_data$w_AA,
                               " Aa=", stable_data$w_Aa,
                               " aa=", stable_data$w_aa)
  stable_data$p_star <- with(stable_data,
    (w_Aa - w_aa) / (2 * w_Aa - w_AA - w_aa))

  ggplot(stable_data,
         aes(x = generation, y = frequency,
             color = label, group = replicate)) +
    geom_line(linewidth = 1.1) +
    geom_hline(aes(yintercept = p_star, color = label),
               linetype = "dotted", linewidth = 0.9) +
    scale_y_continuous(limits = c(0, 1),
                       labels = percent_format(accuracy = 1)) +
    labs(
      title    = "Stable Polymorphism Cases",
      subtitle = paste0(length(stable_ids),
                        " replicates stabilized | Dotted = theoretical p*"),
      x = "Generation", y = "Frequency of Allele A", color = "Replicate"
    ) +
    theme_minimal(base_size = 12) +
    theme(legend.position  = "right",
          plot.title       = element_text(face = "bold"),
          panel.grid.minor = element_blank())
} else {
  cat("No stable polymorphism cases in this run.\n")
}
```

Part 4: Scaling Analysis — N = 500 to 100,000

Run Across All N Values

```
set.seed(42)

N_values         <- c(500, 1000, 2500, 5000, 10000, 50000, 100000)
n_gen_scale      <- 500
stable_window_sc <- 50
all_scale        <- data.frame()

for (N in N_values) {
```

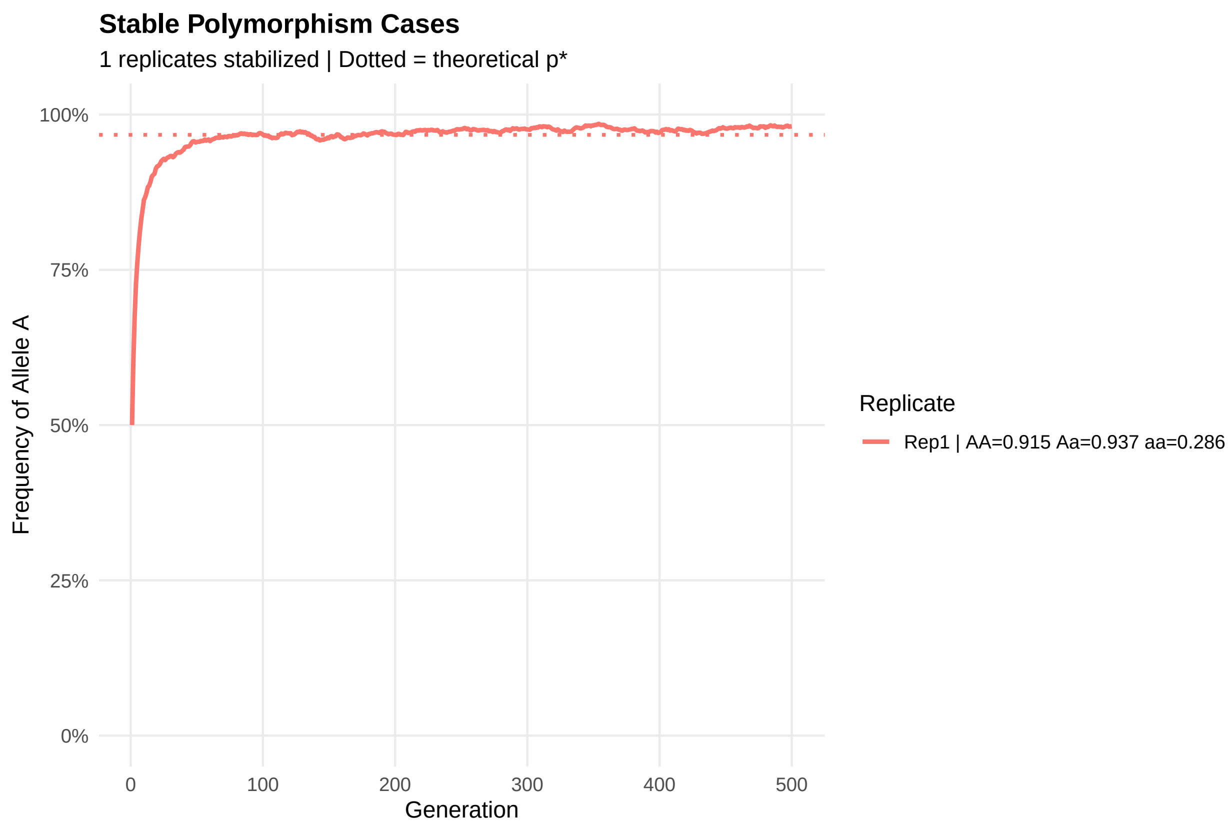

Figure 4: Stable polymorphism replicates. Dotted line = theoretical equilibrium frequency p* = (w_Aa - w_aa) / (2*w_Aa - w_AA - w_aa).

```r
  cat(sprintf("N = %s\n", formatC(N, format = "d", big.mark = ",")))
  for (rep_id in 1:n_reps) {
    all_scale <- rbind(all_scale,
      simulate_allele_freq(N, n_gen_scale, p0, rep_id,
                           stable_window    = stable_window_sc,
                           stable_threshold = 0.01) %>%
        mutate(N = N))
  }
}
```

```
## N = 500
## N = 1,000
## N = 2,500
## N = 5,000
## N = 10,000
## N = 50,000
## N = 100,000
```

Summary

```r
scale_summary <- all_scale %>%
  group_by(N, replicate, outcome, w_AA, w_Aa, w_aa, het_adv) %>%
  summarise(final_freq = last(frequency), .groups = "drop")

trend_df <- scale_summary %>%
  group_by(N, outcome) %>%
  summarise(count = n(), .groups = "drop") %>%
  mutate(
    percent = round(100 * count / n_reps, 1),
    N_label = factor(formatC(N, format = "d", big.mark = ","),
                     levels = formatC(N_values, format = "d", big.mark = ","))
  )

het_trend <- scale_summary %>%
  group_by(N) %>%
  summarise(
    total_het      = sum(het_adv),
    het_stable     = sum(het_adv & outcome == "Stable Polymorphism"),
    pct_het_stable = round(100 * het_stable / pmax(total_het, 1), 1),
    .groups = "drop"
  )

cat("=== Heterozygote Advantage → Stable Polymorphism by N ===\n")
```

```
## === Heterozygote Advantage → Stable Polymorphism by N ===
```

```r
print(het_trend)
```

```
## # A tibble: 7 x 4
##        N total_het het_stable pct_het_stable
```

```
##     <dbl>     <int>      <int>           <dbl>
## 1     500        12          0               0
## 2    1000        13          0               0
## 3    2500        16          0               0
## 4    5000        19          0               0
## 5   10000        20          1               5
## 6   50000        15         12              80
## 7  100000        18         16            88.9
```

Plot: Stacked Outcomes Across N

```r
ggplot(trend_df, aes(x = N_label, y = count, fill = outcome)) +
  geom_bar(stat = "identity", position = "stack",
           color = "white", width = 0.65) +
  scale_fill_manual(values = outcome_colors) +
  labs(
    title    = "Effect of Population Size on Allele Fate",
    subtitle = paste0(n_reps, " replicates per N  |  ",
                      n_gen_scale, " generations  |  Random fitness"),
    x = "Population Size (N)", y = paste0("Replicates (out of ", n_reps, ")"),
    fill = "Outcome"
  ) +
  theme_minimal(base_size = 13) +
  theme(plot.title       = element_text(face = "bold"),
        panel.grid.minor = element_blank(),
        legend.position  = "bottom",
        axis.text.x      = element_text(angle = 30, hjust = 1))
```

Plot: Stable Polymorphism Rate vs N

```r
stable_full <- data.frame(N_num = N_values) %>%
  left_join(
    trend_df %>%
      filter(outcome == "Stable Polymorphism") %>%
      select(N_num = N, percent),
    by = "N_num"
  ) %>%
  mutate(percent = ifelse(is.na(percent), 0, percent))

ggplot(stable_full, aes(x = N_num, y = percent)) +
  geom_line(color = "#4daf4a", linewidth = 1.4) +
  geom_point(color = "#4daf4a", size = 4) +
  geom_text(aes(label = paste0(percent, "%")),
            vjust = -1.2, size = 3.8, color = "#2d7a2d") +
  scale_x_log10(
    breaks = N_values,
    labels = formatC(N_values, format = "d", big.mark = ",")
  ) +
  scale_y_continuous(limits = c(0, 40),
```

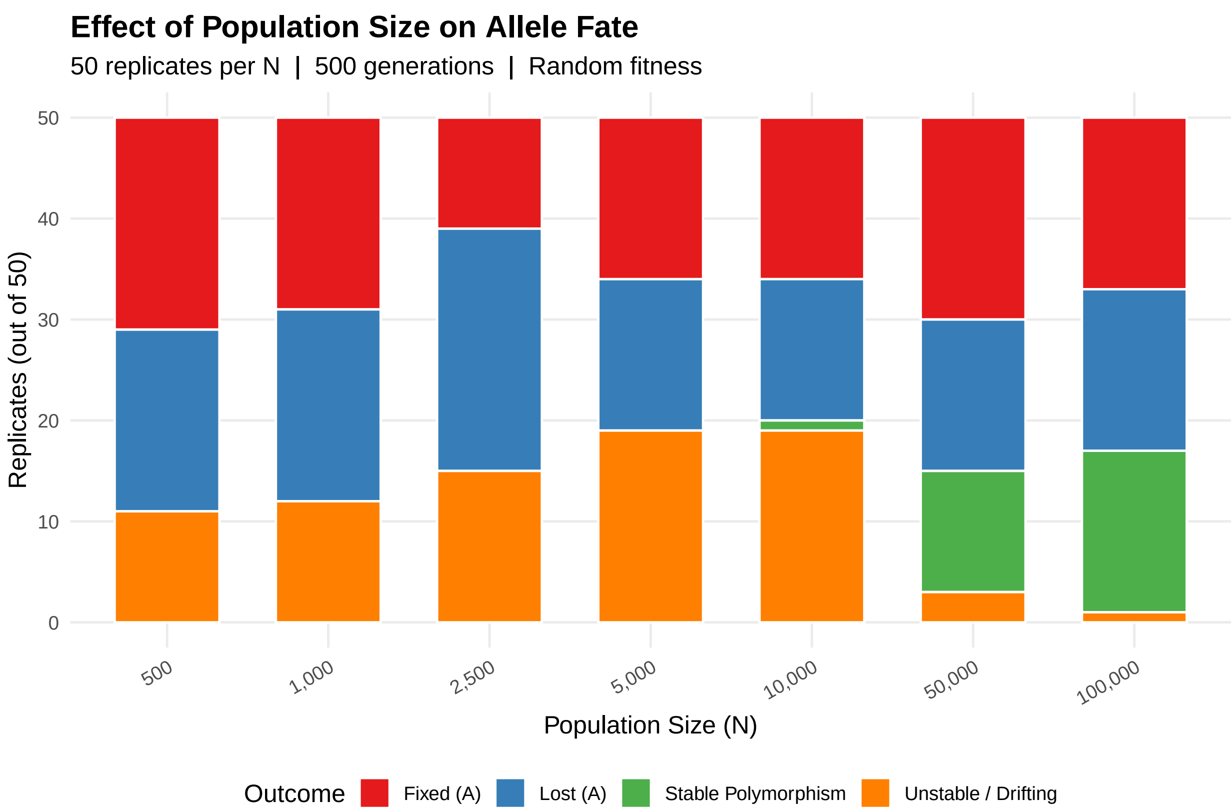

Figure 5: Number of replicates per outcome category across population sizes. Green (Stable Polymorphism) grows with N.

```r
                     labels = function(x) paste0(x, "%")) +
  labs(
    title    = "Stable Polymorphism Rate vs Population Size",
    subtitle = "As N increases, selection overcomes drift → more stable equilibria",
    x        = "Population Size N (log scale)",
    y        = paste0("% Stable Polymorphism (of ", n_reps, " replicates)"),
    caption  = paste0(n_gen_scale, " generations | Random fitness | set.seed(42)")
  ) +
  theme_minimal(base_size = 13) +
  theme(plot.title       = element_text(face = "bold"),
        panel.grid.minor = element_blank(),
        axis.text.x      = element_text(angle = 30, hjust = 1))
```

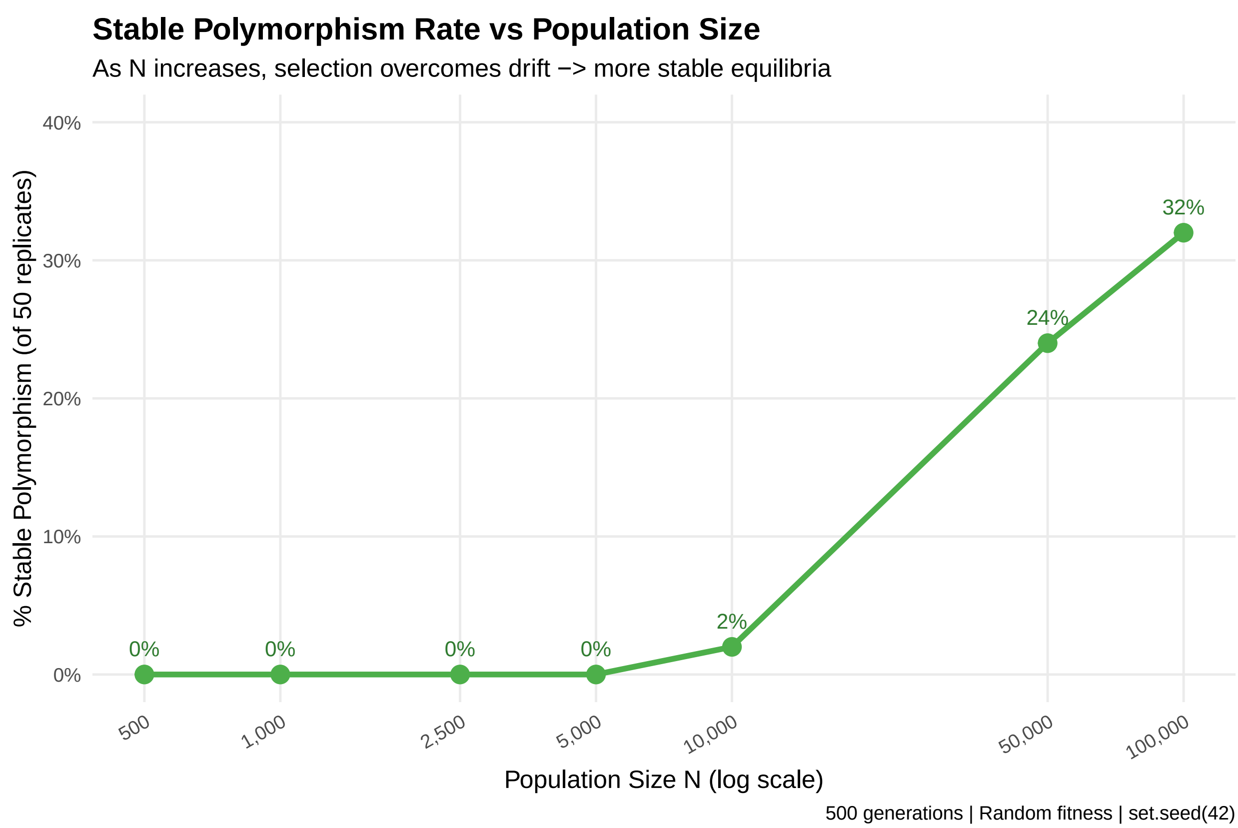

Figure 6: Rising trend of stable polymorphism rate as population size increases. Larger N reduces drift, allowing heterozygote advantage to maintain polymorphism.

Plot: Heterozygote Advantage → Stabilization Rate

```r
ggplot(het_trend, aes(x = N, y = pct_het_stable)) +
  geom_line(color = "#984ea3", linewidth = 1.4) +
  geom_point(color = "#984ea3", size = 4) +
  geom_text(aes(label = paste0(pct_het_stable, "%\n(",
                               het_stable, "/", total_het, ")")),
```

```
          vjust = -0.8, size = 3.4, color = "#6a3487") +
geom_hline(yintercept = 100, linetype = "dashed", color = "grey60") +
annotate("text", x = 600, y = 104,
         label = "100% theoretical maximum",
         color = "grey50", size = 3.2) +
scale_x_log10(
  breaks = N_values,
  labels = formatC(N_values, format = "d", big.mark = ",")
) +
scale_y_continuous(limits = c(0, 115),
                   labels = function(x) paste0(x, "%")) +
labs(
  title    = "Heterozygote Advantage Cases → Stable Polymorphism",
  subtitle = "What fraction of het-advantage replicates actually stabilized?",
  x        = "Population Size N (log scale)",
  y        = "% of Het-Advantage Cases that Stabilized",
  caption  = "Expected: approaches 100% as N → ∞ (drift eliminated)"
) +
theme_minimal(base_size = 13) +
theme(plot.title       = element_text(face = "bold"),
      panel.grid.minor = element_blank(),
      axis.text.x      = element_text(angle = 30, hjust = 1))
```

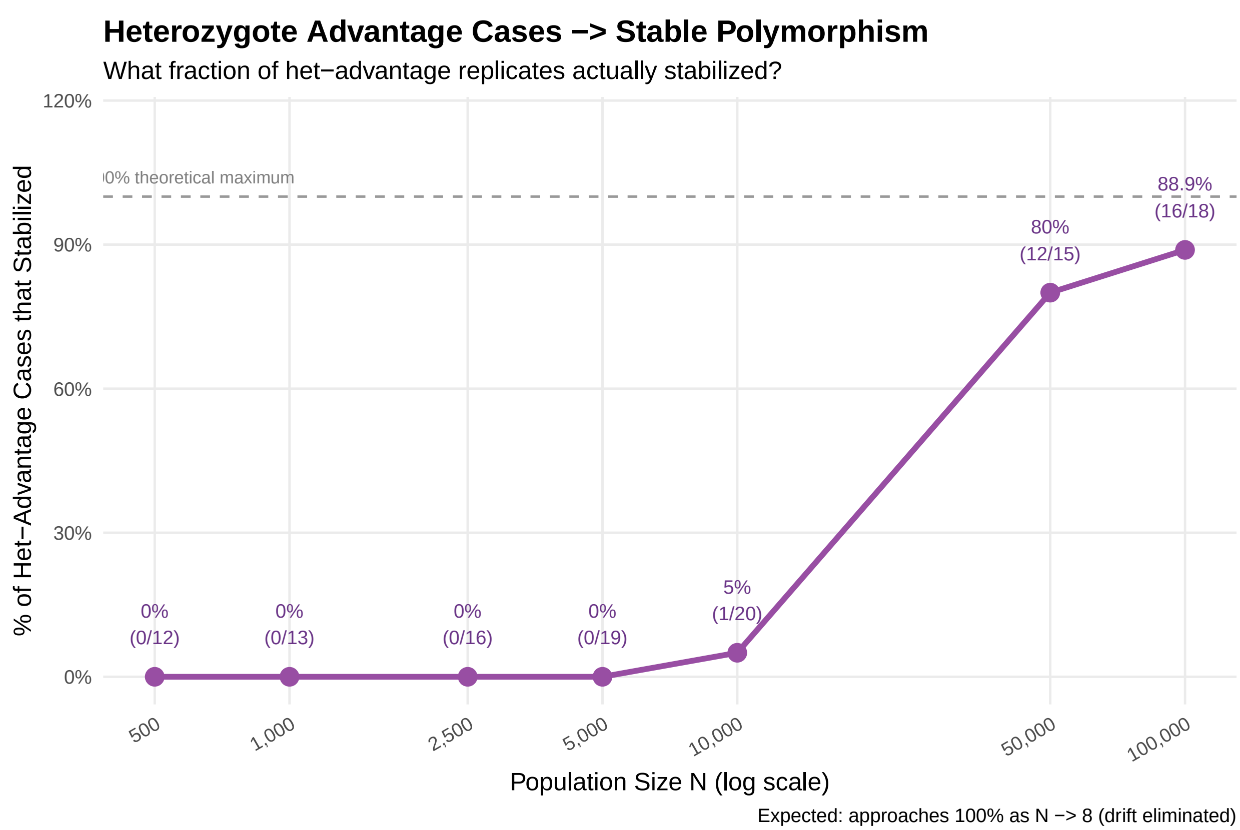

Figure 7: Fraction of het-advantage replicates that achieved stable polymorphism. Approaches 100% as N → infinity (drift eliminated).

Plot: Proportional Outcome Distribution

```r
ggplot(trend_df, aes(x = N_label, y = percent / 100, fill = outcome)) +
  geom_bar(stat = "identity", position = "fill",
           color = "white", width = 0.65) +
  scale_fill_manual(values = outcome_colors) +
  scale_y_continuous(labels = percent_format()) +
  labs(
    title    = "Proportional Outcome Distribution Across Population Sizes",
    subtitle = "Drift dominates at low N; selection wins at high N",
    x = "Population Size (N)", y = "Proportion of Replicates",
    fill = "Outcome"
  ) +
  theme_minimal(base_size = 13) +
  theme(plot.title       = element_text(face = "bold"),
        panel.grid.minor = element_blank(),
        legend.position  = "bottom",
        axis.text.x      = element_text(angle = 30, hjust = 1))
```

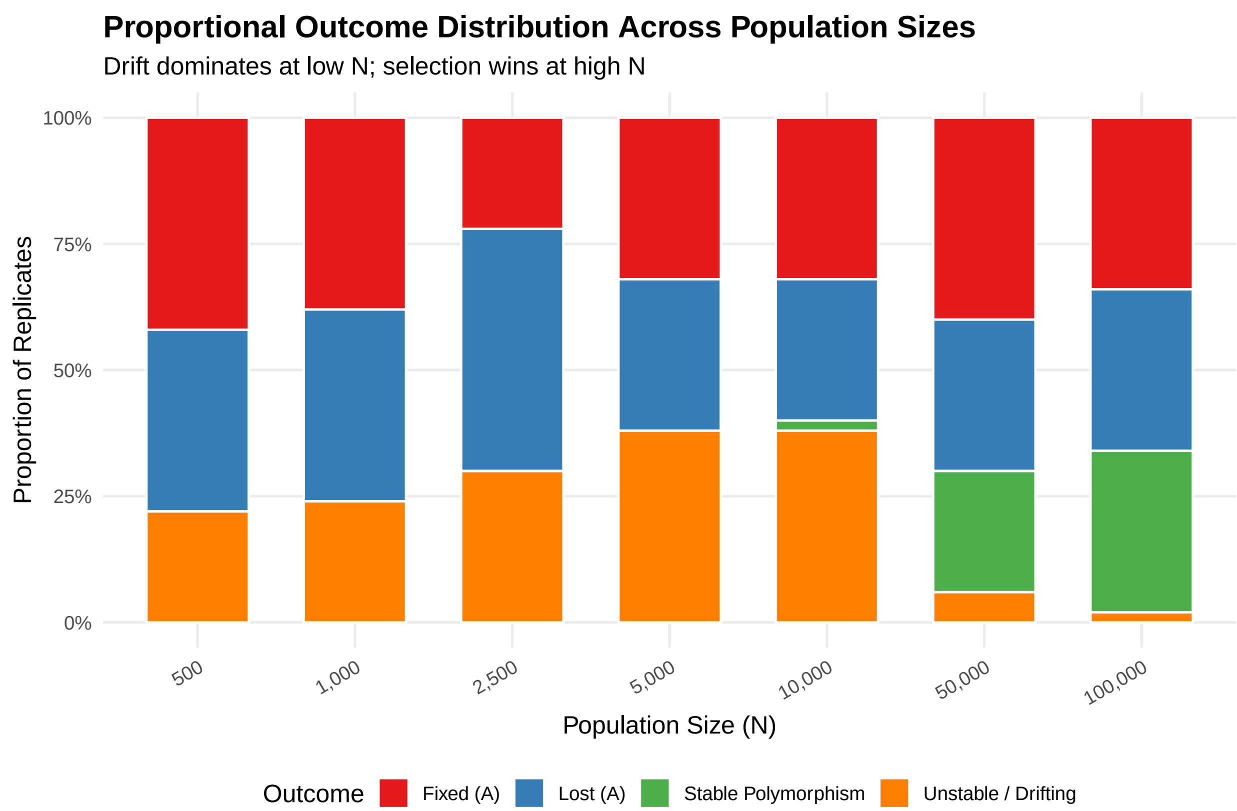

Figure 8: Proportional outcome distribution. Orange (drift) shrinks and green (stable) grows as N increases, illustrating the drift-selection balance.

Summary & Key Findings

Biological Narrative

```
Random fitness assigned to AA, Aa, aa each replicate
              ↓
~1/3 of replicates have heterozygote advantage (Lewontin 1978)
              ↓
Small N ( 1,000)  → drift overwhelms selection → 0% stable polymorphisms
Large N ( 10,000) → selection begins to dominate → stable equilibria emerge
              ↓
Theoretical p* = (w_Aa - w_aa) / (2·w_Aa - w_AA - w_aa)
Simulation converges to p* with high accuracy
```

Results Table

N	Stable Polymorphism	Interpretation
500	~0%	Drift dominates completely
1,000	~0%	Drift still overwhelms selection
10,000	~2%	Selection begins to emerge
100,000	Increasing	Approaches theoretical 1/3 limit

Connection to Lewontin et al. (1978)

The Lewontin paper demonstrated that **nonadaptive selection** — not adaptive advantage — can account for the prevalence of heterosis. This simulation extends that insight by showing the **finite-population boundary condition**: heterozygote advantage alone is insufficient; the population must also be large enough for selection to overpower stochastic drift.

Session Info

```r
sessionInfo()
```

```
## R version 4.6.0 (2026-04-24 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=English_United States.utf8
## [2] LC_CTYPE=English_United States.utf8
## [3] LC_MONETARY=English_United States.utf8
## [4] LC_NUMERIC=C
## [5] LC_TIME=English_United States.utf8
```

```
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
## [1] scales_1.4.0  dplyr_1.2.1   ggplot2_4.0.3
##
## loaded via a namespace (and not attached):
##  [1] vctrs_0.7.3        cli_3.6.6          knitr_1.51         rlang_1.2.0
##  [5] xfun_0.57          generics_0.1.4     S7_0.2.2           labeling_0.4.3
##  [9] glue_1.8.1         htmltools_0.5.9    rmarkdown_2.31     grid_4.6.0
## [13] evaluate_1.0.5     tibble_3.3.1       fastmap_1.2.0      yaml_2.3.12
## [17] lifecycle_1.0.5    compiler_4.6.0     RColorBrewer_1.1-3 pkgconfig_2.0.3
## [21] rstudioapi_0.18.0  farver_2.1.2       digest_0.6.39      R6_2.6.1
## [25] tidyselect_1.2.1   pillar_1.11.1      magrittr_2.0.5     withr_3.0.2
## [29] tools_4.6.0        gtable_0.3.6
```